

Notation:

n^L = number of neurons in layer L. n^0 is number of inputs

w_{ij}^L = weight into layer L, from neuron j to neuron i

W^L = matrix of weights from L-1 to L, dimension 2

b_i^L = bias for neuron i of layer L

B^L = bias vector for layer L

a_i^L = output i of layer L.

A^L = vector output of layer L. A^0 is input vector

z_i^L = weighted sum of neuron i of layer L incl bias.

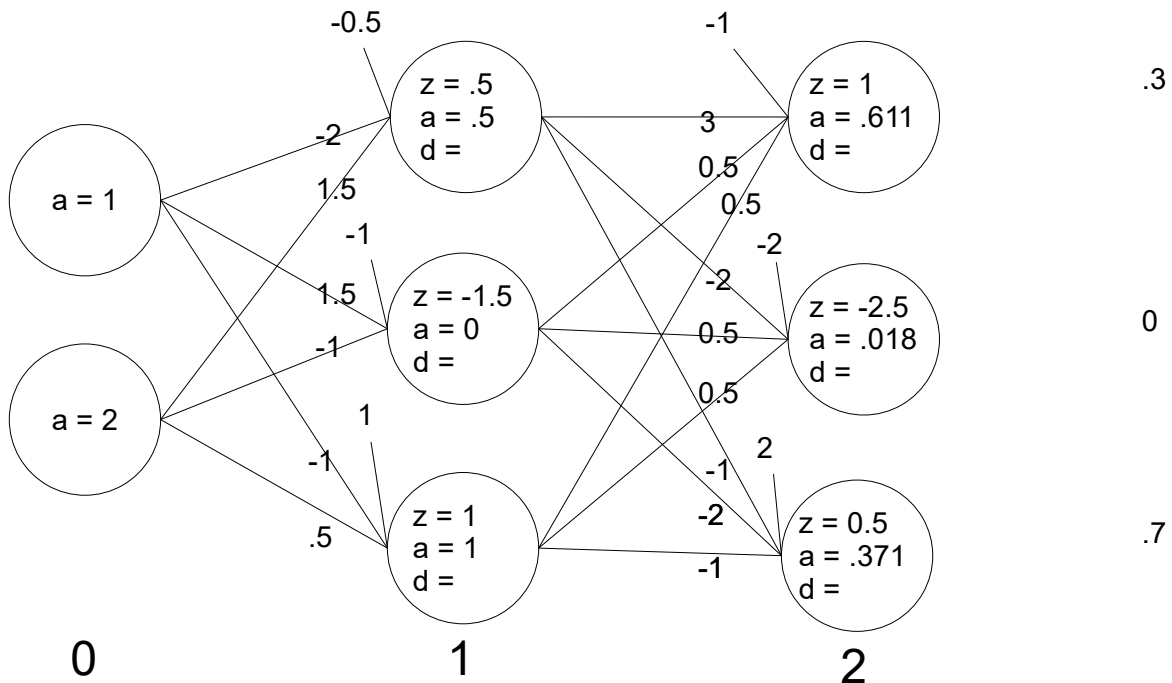
σ = activation function. $a_i^L = \sigma(z_i^L)$, or sometimes, $a_i^L = \sigma(Z^L)$ e.g. for softmax

E = Loss function $E(A^N, Y)$, where N is final layer. (E for "error" to disambiguate from L)

d_i^L = partial of E wrt z_i^L

$$\sigma = \text{relu} = \max(0, z) \quad \sigma = S(z) = 1/(1+e^{-z}) - 1/2 \quad \text{or} \quad \sigma = \text{SM}(z_i) = e^{z_i}/\sum_k e^{z_k} \quad E = \sum_k (a_k - y_k)^2 / n \quad \text{or} \quad E = -(\sum_k y_k \ln(a_k) - \sum_k y_k \ln(y_k))$$

Labels (y)



Chain Rule Partials for BackProp

$$\partial E / \partial w_{ij}^x = a_j^{x-1} * \partial E / \partial z_i^x$$

$$\partial E / \partial b_i^x = \partial E / \partial z_i^x$$

$$\partial E / \partial z_i^1 = \partial a_i^1 / \partial z_i^1 * \sum_k (\partial z_k^2 / \partial a_i^1 * \partial E / \partial z_k^2)$$

$$= \partial a_i^1 / \partial z_i^1 * \sum_k (w_{ki}^2 * d_k^2)$$

$$\partial E / \partial z_j = \partial a_j / \partial z_j * \partial E / \partial a_j$$

Partials for Activations ($\partial a / \partial z$):

For relu: 1 if $a > 0$, 0 otherwise

For $S(z)$: $\partial a / \partial z = e^{-z} / (1 + e^{-z})^2$

For $\text{SM}(Z)$ it's more complex since changing any z affects all a 's. So, $\partial a / \partial z$ needs a matrix of $\partial a_j / \partial z_i$ for all j, i :

for $i = j$: $\partial a_j / \partial z_i = a_i(1 - a_i)$

for $i <> j$: $\partial a_j / \partial z_i = -a_i a_j$

Partials for Loss Functions

For crossentropy: $\partial E / \partial a_j = -y_j / a_j$

For MSE: $\partial E / \partial a_j = 2(a_j - y_j) / n$

$$\begin{vmatrix} \partial E / \partial z_0 \\ \partial E / \partial z_1 \\ \partial E / \partial z_2 \end{vmatrix} = \begin{vmatrix} \partial a_0 / \partial z_0 & \partial a_1 / \partial z_0 & \partial a_2 / \partial z_0 \\ \partial a_0 / \partial z_1 & \partial a_1 / \partial z_1 & \partial a_2 / \partial z_1 \\ \partial a_0 / \partial z_2 & \partial a_1 / \partial z_2 & \partial a_2 / \partial z_2 \end{vmatrix} \begin{vmatrix} \partial E / \partial a_0 \\ \partial E / \partial a_1 \\ \partial E / \partial a_2 \end{vmatrix}$$